

Stall-Triggered Policy Switching for DRL Navigation in Trap Scenarios

Composing complementary policies at runtime to eliminate catastrophic navigation failures

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94%

Scenario avg SR
(+27 vs baseline)

88%

Standard env SR
(matches baseline)

0

Catastrophic
failures (0% cells)

0

Additional
training cost

1 Abstract

DRL navigation achieves high success in open environments but **completely fails (0% SR)** in trap scenarios requiring temporary retreat from the goal.

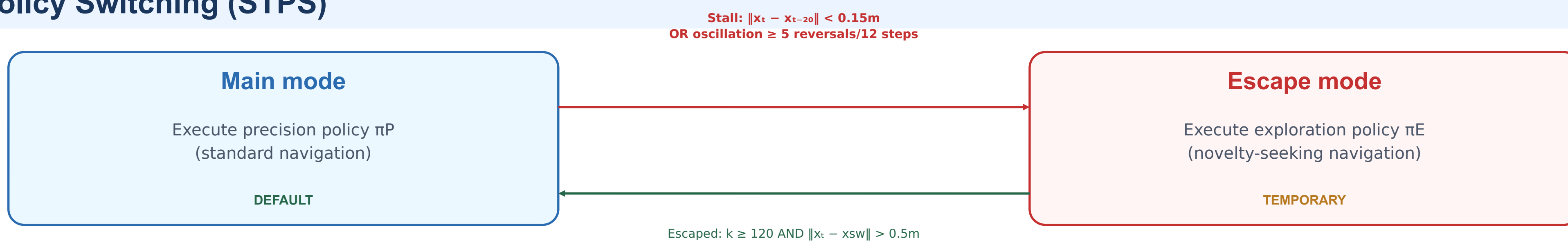
2 Key finding: exploration-precision conflict

Exploration bonus fixes U-trap (0→83%) but **destroys narrow door (100→0%)**. Annealing recovers narrow door but loses U-trap. Irreconcilable.

Method	Std	U-trap	Narrow	Avg
Baseline	87%	0%	100%	67%
Curriculum only	74%	0%	8%	52%
Exploration (πE)	80%	83%	0%	71%
Annealed (πP)	82%	0%	100%	75%

No single policy avoids all 0% cells

3 Method: Stall-Triggered Policy Switching (STPS)



1 Zero training cost
Both policies frozen; only runtime logic

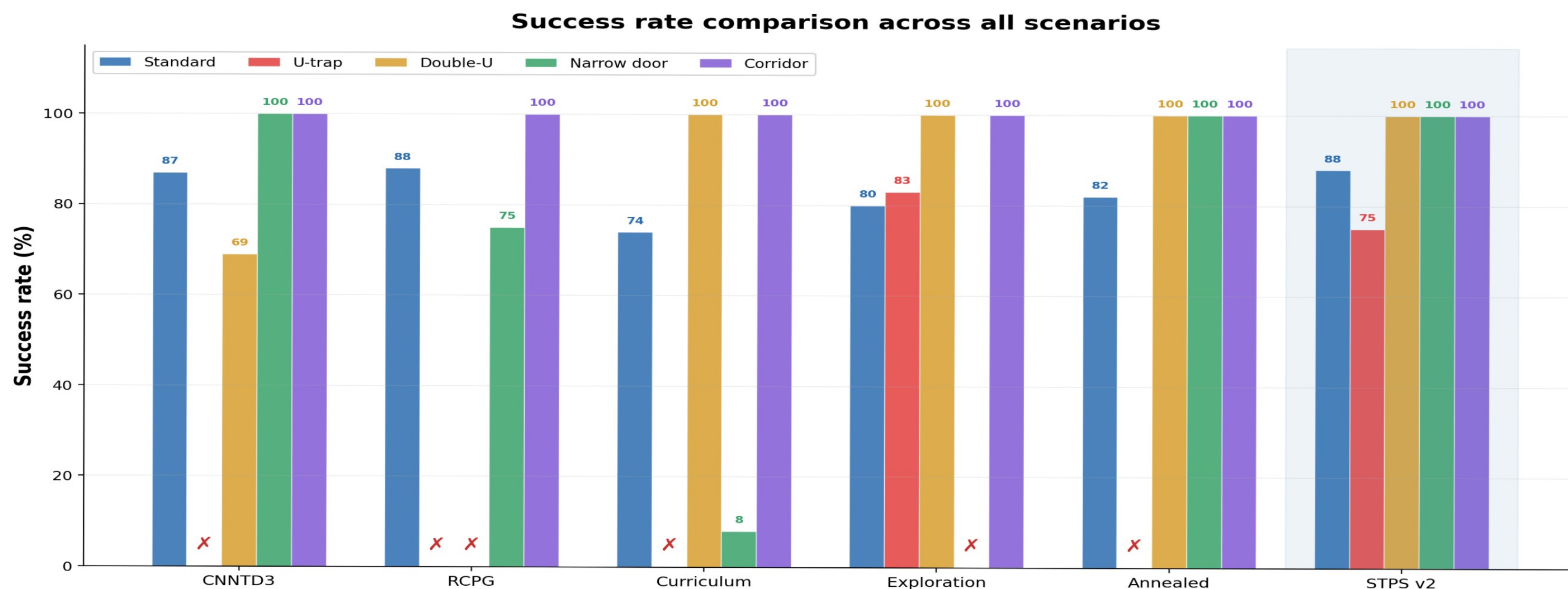
2 Negligible overhead
One distance check per step (<0.1ms)

3 Real-robot ready
Simple if-else, no GPU for switching

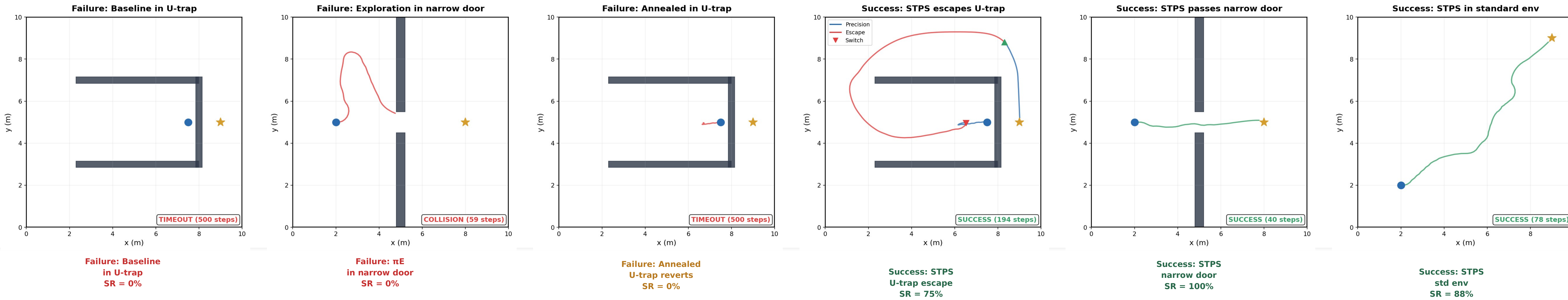
4 Results: STPS outperforms all single policies

Method	Standard (100 ep)	S1 U-trap	S2 Double-U	S3 Narrow door	S5 Corridor	Scenario avg
CNNTD3 baseline	87%	0 ± 0%	69 ± 4%	100 ± 0%	100 ± 0%	67%
RCPG (GRU)	88%	0 ± 0%	0 ± 0%	75 ± 0%	100 ± 0%	44%
Curriculum only	74%	0 ± 0%	100 ± 0%	8 ± 0%	100 ± 0%	52%
Exploration (πE)	80%	83 ± 0%	100 ± 0%	0 ± 0%	100 ± 0%	71%
Annealed (πP)	82%	0 ± 0%	100 ± 0%	100 ± 0%	100 ± 0%	75%
STPS v2 (ours)	88%	75 ± 7%	100 ± 0%	100 ± 0%	100 ± 0%	94%

3 seeds × 12 perturbed starts. STPS is the only method without any 0% catastrophic failure.



5 Success and failure case analysis



6 Parameter sensitivity

U-trap SR (%) / Narrow-door SR (%)
K=120, δ_{prog} =0.5m fixed

	$\delta=0.10$	$\delta=0.15$	$\delta=0.20$
W=15	42/100	42/100	67/100
W=20	67/100	67/100	67/100
W=30	0/100	17/100	25/100

Narrow door = 100% across entire grid
→ switching never harms precision

7 Error taxonomy and what didn't work

Greedy tracking

Cannot retreat — Baseline in U-trap

Reward conflict

Exploration vs precision — πE in narrow door

Parameter space limit

Single network can't encode both — Annealing

Detection latency

20 steps to confirm stall — STPS 75%<83%

What didn't work:

- STPS v3 (goal-distance + progressive escape): no gain over v2
- U-trap specialist: 83%, same as original improved

8 Conclusion

We revealed a hard exploration-precision conflict in DRL navigation and proposed STPS, a training-free switching architecture.

STPS achieves 94% scenario average — the only method without catastrophic failure — while matching baseline on standard environments (88%).

9 Future work

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